

indifferent to green light. Green is also a good night-vision retention colour that allows one to distinguish colours such as yellow and red on maps and charts.

Key component of this flashlight is a 555 timer (IC1) that is configured as an astable multivibrator generating a 50 per cent duty cycle square wave at approximately 6Hz frequency. Three

off switch. In normal mode, IC1 is disabled by normally-closed (N/C) push-to-off switch S1 keeping its output low. As a result, all LEDs (LED1-LED3) glow continuously. On the other hand, when S2 is depressed, IC1 is enabled and all LEDs start flashing at approximately 6Hz frequency.

You can also use this flashlight like a signal lamp. Flashing frequency can

5mm green LEDs (LED1 through LED3) are connected at output pin 3 of IC1. Resistors R1 through R3 limit the current through the LEDs to prevent them from burning up.

Switch S1 is the main on/

be varied by changing the value of capacitor C2.

The entire circuit can be easily assembled on a general-purpose PCB and enclosed in a small plastic box. Drill suitable holes for LEDs and switches. Finally, attach the battery holder using a double-sided glue tape.

EFY note. After looking at a bright light source, you may continue to see an image of that source when you look away. This lingering visual impression is called an afterimage and fades over time. If you use a very bright flashlight, the afterimage overlaps your field of vision and makes it hard for you to see until the afterimage fades. White light containing all frequencies, from red all the way to violet (the visible light spectrum), will ruin your night vision capacity. Red is favoured when you need to recover quickly, while green should be used for acuity. So try using a green light source. ●

HALOGEN LAMP SAVER FOR BIKES

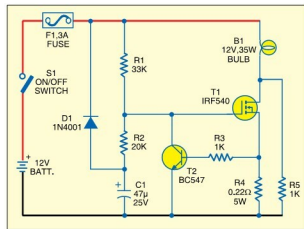
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Halogen lamps are prone to burn-out owing to their low cold current. The rapid heating inside the lamp melts the thin filament and cuts the lamp life short. The circuit described here enhances the life

of the halogen lamp by allowing soft turn-on of the lamp.

When switch S1 is pressed, capacitor C1 charges via resistors R1 and R2, causing MOSFET T1 to conduct. It acts like a normal switch as long as the load (bulb B1) does not draw too much current. Resistor R4 limits the current through the load and also monitors the current flowing through MOSFET T1. The voltage drop across resistor R4 increases as the current through MOSFET T1 increases.

If the voltage at R4 reaches 0.65V, transistor T2 conducts and MOSFET T1 stops conducting. As a result, the lamp stops glowing. Capacitor C1



will hold charge when S1 is switched off. The discharge path for that residual charge is through diode D1 and resistor R5. So the soft turn-on functions correctly any time the bulb is turned on.

Assemble the circuit on a general-purpose PCB and enclose in a small plastic box. Use a heat-sink for mounting the MOSFET in order to dissipate the heat. Current-limiting resistor R4 protects the battery in case the output is short-circuited or the lamp draws more current than specified. Connect the bulb through two external wires having high-current-rating gauge. ●